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LLM-generated content-based explanations for user experience in fashion recommender systems

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Abstract

Recommendation explanations are crucial in helping users make informed and confident decisions, especially in domains such as fashion, where personal style and preferences play an important role. While previous studies have predominantly used review data for explanations, the review-based method requires the availability and quality of a good number of reviews. To address this issue, we investigate the effectiveness of content-based recommendation explanations in fashion recommender systems. Using a Large Language Model (LLM) and deep learning techniques trained on fashion attribute data, we developed a framework that extracts essential visual information from product images and generates user-tailored explanations. This approach allows us to generate customized explanations at various levels—basic, simple, and detailed—for each recommendation. We developed a My Own Style (MOS) interface that displays fashion products, recommendations, and explanations. Our user study with 211 participants showed that detailed explanations, especially when combined with diversity-based algorithms, significantly improved user satisfaction and trust in fashion recommendations. This study contributes to clothing and textile research by providing guidelines for fashion-specific LLM prompts and demonstrating the effectiveness of LLM-generated explanations in fashion e-commerce. Our findings point the way to more personalized and transparent AI-driven fashion recommender systems that improve user experience and style exploration in fashion e-commerce.

Keywords: Fashion, Content-based recommender systems, User experience, XAI, Large language models

Introduction

Recommender systems help us make decisions in many areas of our daily lives and play an important role in the development of personal tastes and preferences. According to preference construction theory (Bettman et al., 1998; Lichtenstein and Slovic, 2006), people tend to form and develop preferences gradually as they go through the decision-making process, rather than being clearly aware of their preferences from the start. The fact that users may not have stable preferences increases the likelihood that they will make decisions based on the information presented by the recommender

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system. Here, it is important to note that, when making a product purchase decision, users with limited previous knowledge may be confined to the information within the recommended results (Chen et al., 2013). This can lead to a filter bubble phenomenon, limiting the scope and diversity of their decisions (Aridor et al., 2020).

Recommender systems in fashion e-commerce are essential systems that not only enhance user satisfaction but also contribute to increased sales. For instance, Stitch Fix reported that by utilizing Artificial Intelligence (AI) to analyze customer preferences and recommend personalized clothing, they secured 3.5 million subscribers and achieved annual revenue of \$2 billion. Since fashion is an important means of expressing individual taste and style, users can more effectively shop for (purchase or explore) fashion products by utilizing the options provided by recommendation systems. However, while these systems effectively tailor product recommendations, users may still experience uncertainty in their decision-making, particularly when they lack transparency in how recommendations are generated. In this context, providing appropriate explanations for recommendation results can reduce users' uncertainty and increase their understanding (Falconnet et al., 2023; Jung et al., 2024; Suvarna and Balakrishna, 2024) and also help them move away from passive information acceptance to active decision-making (Murray and Häubl, 2009; Rosen and Olshavsky, 1987). This necessity for transparent and understandable systems places our research directly within the domain of Explainable AI (XAI), which aims to clarify the reasoning behind AI-driven decisions.

Although previous studies have used product reviews to provide explanations for recommendation results (Chen et al., 2015; Hernandez-Bocanegra and Ziegler, 2021; He et al., 2015; Hernández-Rubio et al., 2019; Hada and Shevade, 2021; Li et al., 2017; Wang et al., 2018), such review-based approaches have several limitations, including the difficulty of generating explanations for products with few or no reviews, or the low quality of reviews (Verma et al., 2022; Yang et al., 2021). In addition, visual elements are often neglected when generating explanations about recommendation results, despite their strong influence on purchase decisions (He and McAuley, 2016; Kang et al., 2017). To address these limitations, this study aims to investigate the effectiveness of explanations generated by large language models (LLMs) in fashion recommender systems. Specifically, we use three types of fashion-related information extracted from user-selected product images: fashion attributes and categories, dominant colors, and visual embeddings to generate recommendations and their explanations. Recommendations are provided through two different algorithms: a similarity-based approach, suggesting products that closely match the visual characteristics of user-selected item, and a diversity-based approach, offering items with moderate similarity but sufficiently diverse styles to encourage exploration. The recommendation results are presented to users through the MOS interface with both product images and corresponding textual explanations, helping users intuitively understand the reasoning behind each recommendation.

However, identifying characteristics of textual and visual information from the product, combining and synthesizing information from two different data modalities, and generating an explanation that is understandable to users and satisfactory is not an easy task. With this motivation, in this paper, we propose a methodology that extracts key information from images of fashion products and uses it to generate explanations through LLMs, one of the state-of-the-art artificial intelligence technologies. LLMs make it possible to

generate explanations by providing relevant information and instructions through carefully designed prompts.

A key consideration in the use of LLMs is the careful design of a prompt to generate effective and reliable explanations. Previous research has highlighted several key factors to build an effective fashion product recommender system (Medeiros Dantas et al., 2021; Dai and Van Hove, 2017; Noh et al., 2023; Obeng et al., 2018). First, to accurately capture the characteristics of fashion products, it is essential to include the information about fashion elements defined by fashion experts, visual information that conveys the shape or mood of the product, and key color information of the product. In addition, when presenting the recommendation results to the user, it is important to consider the accuracy and diversity of the recommendation algorithm (De et al., 2022; Jian et al., 2005; Nurcahya and Supriyanto, 2020; Weihong and Yi, 2006) and the specificity of the recommendation information (Cronin et al., 2000; Yang, 2021). By comprehensively considering these factors, users' understanding and satisfaction with the recommender system can be improved. To this end, we designed LLM prompt structures that include: (1) product information (i.e., fashion attributes, colors, and visual information), (2) recommendation algorithms (i.e., similarity and diversity), and (3) the degree of explanation (i.e., length and amount of information), and generated them through an LLM. To explore the effectiveness of the proposed recommendation explanations and user perception, we developed a web-based fashion recommendation interface called 'My Own Style' (MOS) and conducted a user study with 211 participants.

Through the user study, we found two key insights: (1) Even when the recommendation results are provided by the same algorithm, the level of explanation provided affects users' trust and satisfaction with the results. The more detailed the explanation, the greater the trust and satisfaction with the recommendation. (2) The diversity-based algorithm showed the highest satisfaction when detailed explanations were provided, while the similarity-based algorithm significantly increased user satisfaction when combined with simpler explanations. This suggests that it is crucial to consider the specificity of the explanation according to the algorithm to improve the trust and satisfaction of the recommender system.

Therefore, by developing a framework that generates content-based explanations using an LLM, this study addresses a key research gap left by previous review-based approaches. The contributions of this study are significant in several aspects. We proposed a framework that automatically generates content-based recommendation explanations using LLM, which can compensate for situations where there is insufficient review data. Through a large-scale user study on the proposed content-based recommendation explanations, we confirmed that improved understanding of the recommendation results leads to improved satisfaction and trust in the recommendation results. Furthermore, we discovered that the appropriate type of explanation varies depending on the characteristics of the recommendation algorithm, and based on this finding, we highlighted the importance of generating tailored explanations that take into account the characteristics of the algorithm.

As AI-powered services become more prevalent, users want more trustworthy and satisfying decision-making experiences through additional information about outcomes. This can improve the efficiency of information consumption by supporting meaningful

choices for consumers online. Furthermore, by providing users with opportunities to explore their fashion preferences more deeply, it is possible to foster improvements in self-understanding and an expansion of the diversity of their choices. In this context, our study brings a discussion of opportunities for the use of AI in the fashion domain and directions for effective decision-making support for users' online clothing purchases, contributing to the academic significance of this research in the field of clothing studies. In addition, this study examines the impact of customer-centric recommendation explanations, highlighting the need to consider various stakeholders' perspectives, including vendor requirements. This multifaceted approach suggests future research directions for the effective use of AI and the balanced enhancement of user experience in the fashion e-commerce field.

Literature Review

XAI in recommender systems

With the rapid development of AI, the importance of XAI continues to grow in an effort to increase user understanding and trust in AI, along with interest in AI and human interaction. According to Liao et al. (2022), explainability is essential, especially in popular, user-facing recommender systems on platforms such as e-commerce. Explainable recommender systems are algorithms that provide not only recommendation results but also explanations for why certain items are recommended. These explanations, which can be delivered to users or system designers, help to improve trust and user satisfaction (Zhang et al., 2020).

There are two types of explainable recommendation models: the first is model-intrinsic in terms of interpretability, which makes the model's computation process and decision mechanism transparent enough to provide a natural explanation for the model's decision. The second is a post-hoc approach to generate an explanation after the system makes a decision. Several studies have been conducted to understand how users interact with these explanations. In this study, interaction does not refer to the iterative feedback processes between users and recommender systems, but rather to the cognitive process by which users read textual explanations, understand the rationale behind the recommendations, and subsequently refine their fashion preferences.

Some researchers have tried to develop a better model or build an appropriate mental model for the system by providing a transparent basis for model decisions to domain experts or system developers. There are visual types, such as diagrams, and textual types, such as natural language, in the post-hoc explanations used in previous studies. Zhang et al. (2022) identified users' perceptions of service recommendations for laypeople and confirmed that the textual explanation was the most transparent and satisfying. Previous research emphasized that recommendation satisfaction, trust, explanation transparency, and explanation satisfaction are critical factors for evaluating users' perceptions of recommender systems (Balog and Radlinski, 2020; Chen et al., 2019; Dominguez et al., 2019).

Considering these factors helps clarify how explanations influence users' acceptance and decision-making. These aspects have been repeatedly highlighted because they directly reflect users' cognitive and emotional responses, which are essential to maintaining engagement and ensuring long-term acceptance of recommendations.

Thus, focusing on these dimensions allows for a nuanced understanding of user perceptions, beyond mere accuracy of recommendations. Therefore, we want to understand the interaction between the user and recommendation explanation through a post-hoc, textual explanation. Through this interaction, we aim to support users' active understanding and decision-making in fashion recommender systems.

Preference construction theory in recommendation contexts

In the context of recommender systems, preference construction theory suggests that user preferences are not fixed tastes to be detected but are dynamically constructed when users face choice situations (Payne et al., 1993). Particularly in domains such as fashion, users often interact with systems without clearly defining their preferred styles, making it likely that the recommendations presented by the system significantly influence the formation of users' initial preferences (Lichtenstein and Slovic, 1971; Tversky and Kahneman, 1974). In the early stages of interaction, users tend to rely more heavily on recommendations due to uncertainty, and cognitive psychological mechanisms come into play in this process. For instance, items or style information provided by recommendation systems actively shape users' actual preferences through mechanisms such as the anchoring effect (i.e., a cognitive bias causing users to overly rely on initial information, influencing subsequent judgments) (Tversky and Kahneman, 1974), and this forms users' judgment criteria; the framing effect, which guides users' preference direction by highlighting certain styles; and the position bias, where items positioned at the top of product lists are perceived as more attractive (Joachims et al., 2017). Furthermore, due to the social nature of fashion products, popular items or social proof information provided by the system may exert an even stronger influence on users' preference decisions (Salganik et al., 2006). From this theoretical perspective, it is necessary to recognize that recommendation systems go beyond simply providing information and actively intervene in the process of forming users' tastes and styles. Based on this, we aim to investigate how the explanations provided by recommendation systems affect the process of user preference formation.

Various types in recommender system algorithms

Recommendation algorithms can be categorized into three main approaches: content-based filtering, collaborative filtering, and hybrid methods that combine these two. First, content-based filtering uses product attributes such as images, descriptions, and metadata to recommend products that are similar to users' previous preferences (Kang et al., 2017; Lops et al., 2011). Recent studies in the fashion domain have actively employed deep learning techniques, especially convolutional neural networks (CNNs), to extract visual attributes (e.g., colors, styles, patterns) from product images to provide more nuanced and accurate recommendations (He and McAuley, 2016; Kang et al., 2017).

Second, collaborative filtering leverages similarities between users or items based on historical interaction data (e.g., purchases, ratings, clicks) to provide personalized recommendations (He et al., 2017). However, collaborative filtering approaches face inherent limitations such as data sparsity and the cold-start problem, making them less effective when dealing with new or infrequently interacted products (Schein et al., 2002).

To overcome these problems, hybrid recommendation approaches combine content-based and collaborative filtering, aiming to leverage the complementary strengths of each approach (Burke, 2002; Li et al., 2017). Recent hybrid recommender systems in fashion have extensively employed deep learning techniques to extract visual and textual features for better personalization and style compatibility (Chen et al., 2019).

However, these approaches still face challenges related to data quality, accurately capturing rapidly changing fashion trends, and balancing personalized preferences with overall style consistency (He and McAuley, 2016; Lu et al., 2023). Early recommender systems primarily centered on accuracy by selecting and recommending items that users were likely to prefer from a broad range of information sources (Herlocker et al., 1999; Rashid et al., 2002). While recommending products closely aligned with a user's previous choices can indeed enhance trust and satisfaction, relying exclusively on this similarity-based approach can inadvertently limit users to an increasingly narrow set of options—often referred to as the filter bubble problem¹ (Ge et al., 2010; Mcnee, 2006).

Consequently, diversity has emerged as a critical factor in modern recommender system design to counteract this narrowing effect. By broadening the scope of recommendations, diversity helps expand user exposure beyond items that merely mirror past preferences, thereby mitigating the pitfalls of over-personalization (De Pessemier et al., 2014; Herlocker et al., 2004; Taramigkou et al., 2013). Building upon these insights, our work highlights how balancing similarity with diversity can enhance user satisfaction more effectively than accuracy alone, ensuring that users receive both relevant and varied recommendations. To address these limitations, our study adopts a content-based filtering approach using visual and textual attributes extracted directly from fashion products. Specifically, we aim to investigate how recommendation explanations can effectively support user decision-making in scenarios where users have not yet clearly established their preferences or lack sufficient interaction history with the recommender system.

Types of explanation approaches in recommender systems

Because content-based filtering calculates recommendation results based on product or content features, content-based explanations that use these features can generate intuitive explanations based on the domains, contextual situations of recommendations, and extracted features (Gao and Zhou, 2022). Several studies have used user reviews to improve the quality of recommendation models and explanations (Chen et al., 2015; Hernandez-Bocanegra and Ziegler, 2021; He et al., 2015; Hernández-Rubio et al., 2019; Hada and Shevade, 2021; Li et al., 2017; Wang et al., 2018). Zhang et al. (2022) extracted keywords from reviews in a restaurant recommendation domain and created explanations using features, such as distance, price, and taste, and adjusted the number of features included in the explanation (i.e., two, three, and four features) to confirm user preference. Users had the highest trust and satisfaction when they were presented with an explanation containing three or four features. Reviews based on a user's subjective

¹ The filter bubble problem refers to a situation in which personalization algorithms isolate users in their own informational bubbles by continuously recommending similar content based on past behavior. This limits one's exposure to diverse perspectives and potentially reinforce existing preferences while preventing new discoveries (Pariser, 2011).

experience have the advantage of including both positive and negative aspects, which can help other users make reasonable decisions (Zhang et al., 2022).

In the fashion domain, Chen et al. (2019) proposed a recommender system that generates detailed textual explanations by extracting specific fashion attributes such as style, fabric quality, and fit directly from user-generated reviews. Their study demonstrated that explanations derived from reviews significantly improved user understanding and satisfaction in fashion recommendation contexts.

However, review-based explanations face limitations as they cannot be generated without reviews, and the quality of the explanations is directly dependent on the reviews (Verma et al., 2022). These limitations can also affect the quality of the explanation and contribute to popularity bias, where only popular products are exposed (Ciampaglia et al., 2018). Yang et al. (2021) confirmed that attribute-level explanations encouraged users to compare recommendations, which enhanced their understanding and confidence in the recommended products. For these reasons, we aimed to design an content-based explanation that (1) can be generated for all products, (2) is of comparable quality for all products, and (3) can extract keywords using the product's information.

To this end, we extracted three types of information from fashion products using deep learning models, using them as an input to generate content-based explanations of large-scale data using the LLMs. In addition, although the explanations may change depending on the type of algorithm, most research has only verified user perceptions for a single algorithm. Therefore, we wanted to investigate the difference in explanations generated by the two algorithms, similarity and diversity on user perceptions.

Large language models for generating recommendation explanations in e-commerce

In e-commerce, there are a large number of products, and new products are added every day. Therefore, considering the real-world e-commerce situation, much consideration is needed to quickly and accurately generate explanations that reflect product characteristics, recommendation situations, and algorithms. Existing studies that have confirmed user perceptions by generating recommendation explanations have attempted to generate natural language explanations directly through crowd workers (Lu et al., 2023; Millicamp et al., 2019) or learning models from large-scale language data (Verma et al., 2022). However, manual explanation generation comes with a high cost and does not always guarantee good results. Pre-trained models can be domain-biased depending on the learned data, and learning takes a lot of time and computational resources to reflect all the complex requirements of the real-world recommendation situation.

To overcome these challenges, Balog et al. (2023) generated explanations by an LLM pre-trained with large-scale language data based on the keywords extracted from reviews. However, the paper only generated explanations for a single ensemble algorithm (i.e., an ensemble of three machine-learning algorithms: k-nearest neighbors, Weighted regularized matrix, and sparse linear method). Regarding the two types of algorithms, we aimed to conduct experiments on prompt engineering, which is appropriate for e-commerce fashion recommendation situations, and to explain the product information and inherent meaning of the algorithms. To the best of our knowledge, ours is the first study to use information extracted from products for both algorithm development and

explanation generation and to substantiate user perceptions by attempting to generate content-based explanations through an LLMs.

Comparison of previous studies

Previous studies have used largely review-based explanations and single recommendation algorithms, often overlooking the significance of visual attributes and tailoring explanation styles to different algorithms. Table 1 systematically compares recent studies on explainable recommender systems, highlighting key differences in the domain of application, explanation generation methods, data sources, and user evaluation practices. As presented in Table 1, most previous studies used rule-based explanation approaches, such as keyword extraction from reviews, ranking-based explanations, or manual explanations created by experts. Additionally, these studies typically evaluated explanations within a single recommendation algorithm, without addressing how

Table 1 Comparison with previous studies on explanation generation in recommender systems

Title	Domain	Used algorithm	Explanation data source	Explanation generation method
Let Me Explain: Impact of Personal and Impersonal Explanations on Trust in Recommender Systems (Kunkel et al., 2019)	Movie recommendation	Single (collaborative filtering)	User reviews	Human, Rule-based
To Explain or Not to Explain: The Effects of Personal Characteristics when Explaining Music Recommendations (Millecamp et al., 2019)	Music recommendation	Single (hybrid)	User history	Rule-based
Explaining Review-Based Recommendations: Effects of Profile Transparency, Presentation Style and User Characteristics (Hernandez-Bocanegra and Ziegler, 2021)	Hotel recommendation	Single (collaborative filtering)	User reviews	Rule-based
Effects of Feature-Based Explanation and Its Output Modality on User Satisfaction with Service Recommender Systems (Zhang et al., 2022)	Restaurant recommendation	Single (content-based filtering)	Item metadata (i.e., price, location)	Rule-based
User Perception of Recommendation Explanation: Are Your Explanations What Users Need? (Lu et al., 2023)	Video/Movie recommendation	Single (hybrid)	User profile	Human, Rule-based
Our approach	Fashion recommendation	Multiple (content-based filtering)	Product data (i.e., category, attribute, color)	LLM-based

Previous studies mostly focused on evaluating user perceptions by generating explanations based on a single recommendation algorithm. In contrast, our approach explores the effectiveness of explanations for two different recommendation algorithms. Additionally, we utilize visual product data as a source to address the limitations of review-based approaches

explanations might vary across algorithms. Our research distinguishes itself by introducing an automated approach using an LLM to generate explanations directly from visual product attributes and metadata. Furthermore, we uniquely compare the effectiveness of explanations across two recommendation algorithms—similarity-based and diversity-based—allowing us to identify optimal explanation types for each algorithm. This approach aims to enhance user understanding, trust, and satisfaction by providing personalized and algorithmically tailored explanations.

Methods

To investigate user perception of recommendations, we developed a fashion recommendation web interface, My Own Style (MOS). To conduct a user study, we collected data on fashion products from e-commerce websites and used deep learning models to extract information about the product's style, fashion attributes, and color. We designed two types of recommendation algorithms and three types of explanations. Figure 1 illustrates our recommendation process, and Fig. 2 shows our web interface for representing the recommendation results.

Dataset

We collected fashion product data from e-commerce sites through web crawling. Between March and April 2023, we gathered a total of 13,395 product images, titles, and prices from the world's largest fashion e-commerce sites (H&M,² COS,³ and ZARA⁴). To provide participants with a broader and more realistic selection of fashion products, we also used approximately 105,000 items from the H&M Personalized Fashion Recommendation Challenge dataset provided by Kaggle (Ling et al., 2022). This dataset, originally designed for developing personalized recommendation algorithms, includes product metadata, images, and transaction histories, allowing for diverse and realistic recommendation scenarios. Two researchers reviewed the collected data in two steps. First, we automatically removed products that were excluded from the study, such as bags, shoes, accessories, and baby clothes. In the second step, duplicate images within the dataset were manually removed by the researchers. As a result, 73,355 fashion products were used in the user study.

Extracting information from fashion product

We systematically extracted visual and textual information from fashion product images through three distinct methods, with the detailed process and procedures illustrated in Fig. 3. Visual information regarding the appearance of products (e.g., style, overall shape) is one of the important factors influencing consumers' decisions to purchase clothing (Dai and Van Hove, 2017). To extract visual information from products, we employed Vision Transformer-Base (ViT-B) (Dosovitskiy et al., 2020), commonly used for image embedding, based on previous research. By inputting product images into

² <https://www.hm.com/>

³ <https://www.cos.com/>

⁴ <https://www.zara.com/>

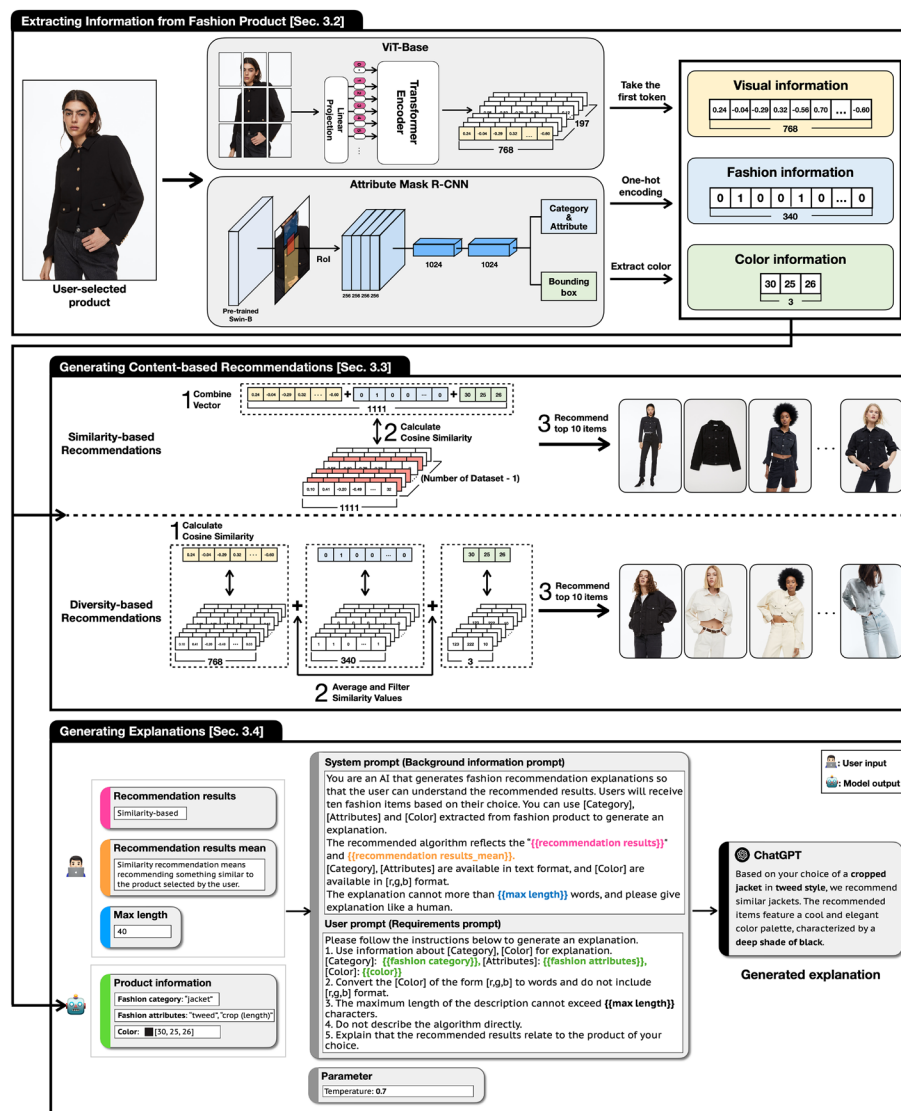


Fig. 1 Process of developing recommendation algorithms and generating explanations of a product selected by the user. Section [Extracting information from fashion product](#) shows how to extract three types of information from a product using two deep-learning models. Section [Generating content-based recommendations](#) shows the method for two different recommendations (similarity-based and diversity-based). Section [Generating explanations](#) shows the steps to generate an explanation. We used GPT-3.5-turbo to generate our content-based explanations

ViT-B, we utilized the last layer's hidden state that contains 768-dimensional information of the entire image as the visual information. This information comprehensively includes overall shape, style, and pattern of the product image.

Fashion consists of two distinct elements: categories and attributes (Jia et al., 2018). Categories represent the fundamental parts of a garment (e.g., "shirts," "pants," and "coats"). Attributes are specific characteristics that offer more detailed explanations within these categories (e.g., "v-neck" and "loose fit"). This information is crucial in content-based recommender systems for accurately identifying the characteristics of fashion

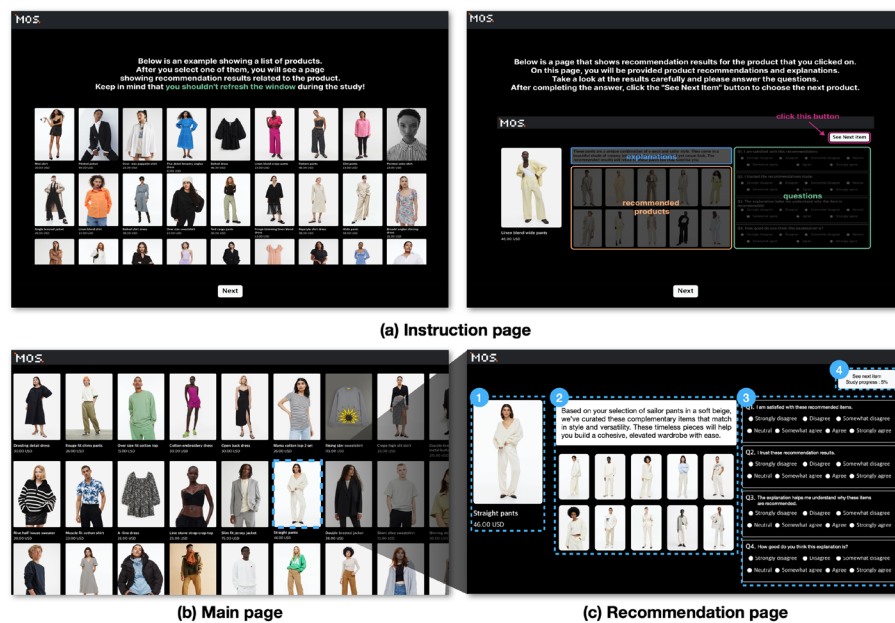


Fig. 2 Screenshots of the MOS interface we developed for user study: **a** shows the instruction page, which guides participants on how to proceed with the study; **b** shows the main page where a variety of products are displayed for selection; and **c** illustrates the recommendation page, where information about the selected product is displayed. On the recommendation page, ① is information about the product selected by a participant, ② shows the recommendation results of our algorithm and explanation, ③ refers to the dependent variable questions, and ④ is the progress rate of the study

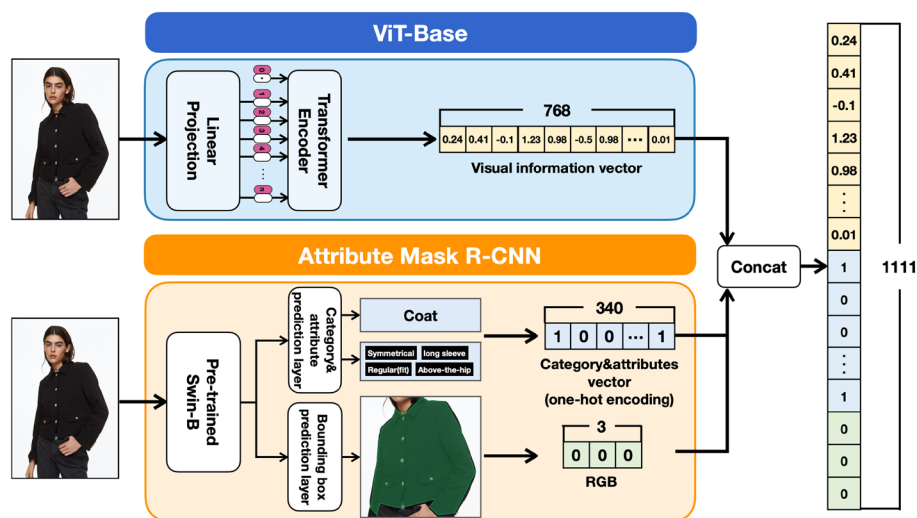


Fig. 3 Process of extracting three types of information from a single product image using two deep-learning models. The ViT-Base model captures the product's overall shape and style as a high-dimensional visual embedding, and the Attribute Mask R-CNN extracts category, attribute, and color information. Finally, these representations are concatenated into a unified vector for downstream recommendation and explanation generation

products and offering personalized recommendations that align with users' individual styles and preferences (Noh et al., 2023). We utilized 46 categories and 294 attributes

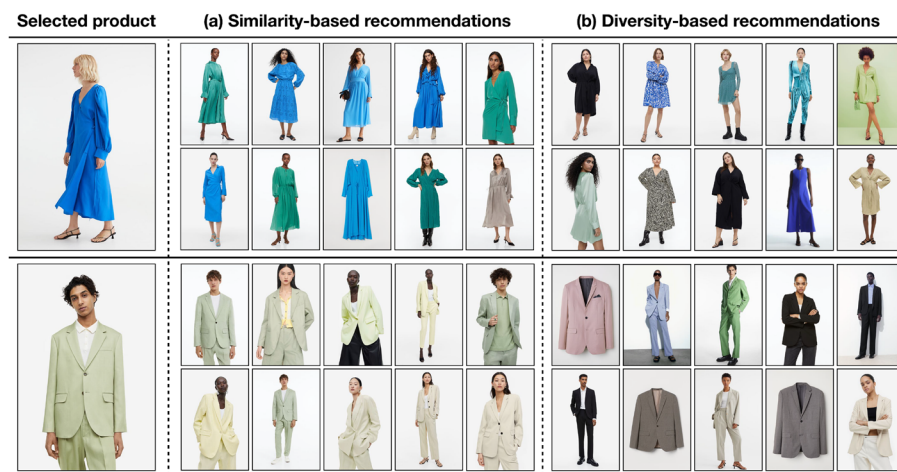


Fig. 4 Results of two recommendation algorithms: The left images show selected products (baseline image). **a** shows the recommendation results of the similarity-based algorithm, and **b** shows the results of the diversity-based algorithm

defined by fashion experts in Fashionpedia (Jia et al., 2020) based on previous research in the fashion domain. We employed the Attribute Mask R-CNN to detect the attributes and categories. As a result, we used one-hot embedding to represent these elements in a 340-dimensional vector.

Color in fashion products contributes to enhancing their value (Medeiros Dantas et al., 2021) and is a factor that positively influences users in identifying and selecting products (Obeng et al., 2018). Since Fashionpedia does not specify product colors, we decided to use the bounding box corresponding to the main category identified by the Attribute Mask R-CNN. We used an algorithm⁵ to extract the color with the highest proportion within the largest detected bounding box and embedded it as a 3-dimensional RGB vector.

In this study, we provided extracted information to an LLM, allowing the model to automatically generate a natural language description of the product's style. This method enabled the provision of easy-to-understand style descriptions for users.

Generating content-based recommendations

In this paper, we implemented two recommendation algorithms—similarity and diversity—which are extensively studied in the content-based recommendation. For similarity-based recommendation, we combined three embedding vectors and calculated the cosine similarity based on the previous study (Jeon et al., 2021). For the diversity-based recommendation, we first calculate the cosine similarity between the selected product and other products using three different types of information. After that, we average and filter the similarity values across all three types of data. Finally, we recommend the top 10 items based on these processed similarity values. We avoided using similarity values that were too high, as they did not reflect diversity. In this work, we only used cases

⁵ <https://github.com/fengsp/color-thief-py>.

where the average similarity was between 0.6 and 0.8 for diversity-based recommendations. Figure 4 shows the results of our recommendations.

Generating explanations

To measure the difference in user satisfaction between conventional and content-based explanations of recommended results in fashion e-commerce, we selected two conventional explanations, as shown in Fig. 5. Then, we designed two levels of content-based explanations to examine how the length of the explanations and the amount of information included affect user perceptions. Figure 1 shows the framework for generating explanations. We integrated product information extracted from Sect. 3.2 and the context and method of recommendations into the prompts, and obtained natural and consistent explanations through prompt engineering. Therefore, the generated explanations include information about the characteristics of the recommendation algorithm and details about the recommended products. We used GPT-3.5-turbo API to generate recommendation explanations. We referred to the conventional explanations as “basic” and the two levels of content-based explanations as “simple” and “detailed,” respectively.

Prompting engineering

Prompt engineering is the process of optimizing the prompts entered into the LLM through various experiments in order to obtain more accurate and relevant responses. In this study, the structure of the prompts and the information used were experimented with and qualitatively evaluated by the researcher.

We designed two types of prompts to generate explanations: (1) system prompts that contain background information to help LLMs better understand the request to generate explanations, and (2) user prompts that contain more specific and clear instructions and information about the products needed to generate explanations.

1. System prompt (background information prompt): Our user study consists of two types of recommendation algorithms. To enable the LLM to understand the shopping scenario, we provided the context of recommending products on e-commerce sites based on the item selected by the user. We assigned the role of generating explanations for the recommendation results. We explained what similar and diverse recommendation results mean. Finally, we instructed that the characteristics of the explanations should be human-like and natural, based on previous research showing that human-authored explanations improved user satisfaction (Lu et al., 2023).
2. User prompt (requirements prompt): After setting up the system prompt, we included specific requirements and the information needed to make the

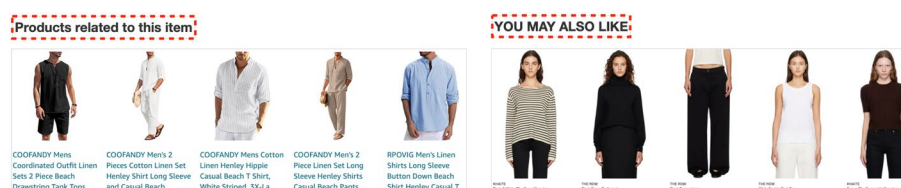


Fig. 5 Examples of conventional recommendation explanations from real-world e-commerce websites. Most websites do not provide information about why items are recommended

recommendation in the prompt. These requirements included instructions to generate an explanation under certain conditions (e.g., the algorithm process, the information extracted from the product, and the desired length of the explanation).

We designed two levels of explanation detail (i.e., simple and detailed) to include different amounts of information. The simple level uses only the product's category and color information, while the detailed level includes additional attribute information to provide a more comprehensive explanation.

To control the length of explanations, we introduced a 'max length' variable in the prompt. To determine the optimal value for this variable, we qualitatively evaluated generated explanations while increasing the value from 10 to 50. The analysis showed that fewer than 15 words resulted in incomplete sentences, while more than 40 words tended to include unnecessary details or reduce compatibility with the e-commerce UI. Based on these results, we set the 'max length' to 15 for simple explanations and 40 for detailed explanations. All prompts and examples of the generated explanations are presented in Fig. 6.

Participants

We recruited participants on Amazon Mechanical Turk (MTurk⁶), which allows us to recruit cloud workers online. We posted a Human Intelligence Task (HIT) on the platform, with participation criteria requiring workers to have (1) a HIT approval rate over 95%, (2) over 500 HITs have been approved, and (3) have purchased clothing online. This study was approved by the author's Institutional Review Board (IRB). According to the research ethics advised by the IRB, we offered participants an informed consent page that explained the research objectives, procedures, duration, compensation, and potential benefits or risks. Only those who agreed to the study could participate. We attempted to ensure data quality by verifying redundant response patterns (e.g., collectively selecting only one response option), which resulted in the removal of nine participants, leaving a total of 211 participants. Power analysis was performed with an alpha of 0.05, a power of 0.8, and a medium effect size ($f = 0.25$), calculated using G*Power. The effect sample size with six conditions and three degrees of freedom is 179, so our experiment had sufficient statistical power for the study. Participants completed the task in an average of 17.5 min and received \$3.0 for their participation.

Study design

In this work, we developed a MOS interface for the user study (Fig. 2). We developed the MOS web page by referencing the interface of a real online shopping mall to provide participants with a realistic e-commerce experience as much as possible. Before entering the experiment, a participant was provided with brief online instructions and directions about the experimental procedure (Fig. 2a). The procedure for each trial consistently followed three steps: (1) after the instructions, the participant selected a preferred product from the main page (Fig. 2b); (2) they were then directed to a new page to view the recommendation results along with the corresponding explanation (Fig. 2c); and (3)

⁶ <https://www.mturk.com/>

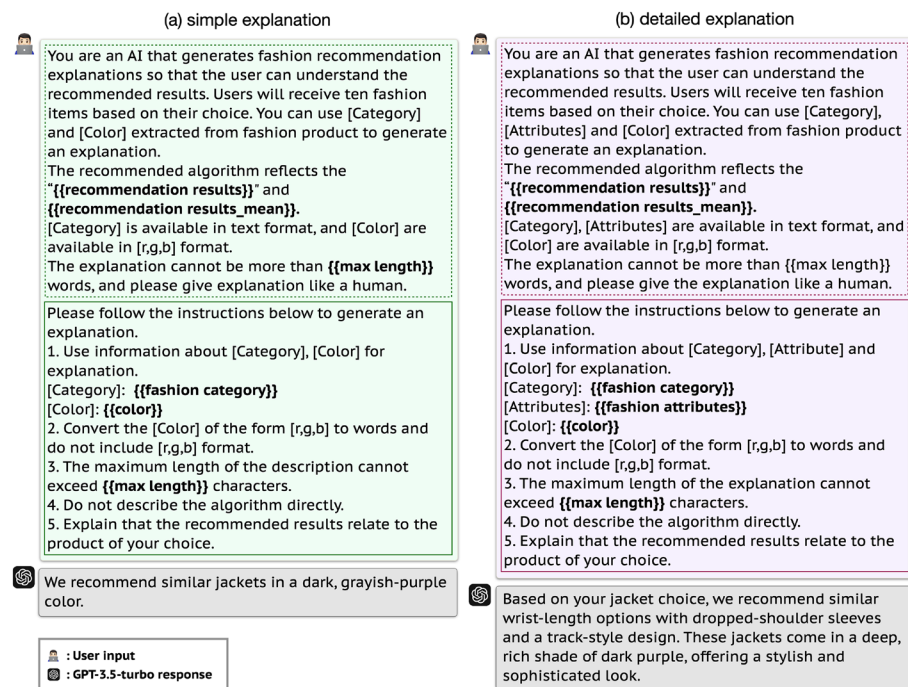


Fig. 6 Examples of the explanation generation process in our recommender system. The figure illustrates **a** a simple explanation generated from a prompt containing only category and color information, and **b** a detailed explanation generated from a prompt that additionally includes specific fashion attributes. These prompts are provided to an LLM to automatically generate textual explanations. In the actual interface, users view these generated explanations along with images of recommended products, allowing them to intuitively understand the rationale behind each recommendation

they immediately answered the survey questions for that instance before proceeding to the next trial.

This study employed a within-subjects design in which each participant was exposed to a total of 18 recommendation instances. Specifically, each participant experienced all six experimental conditions, derived from two types of algorithms and three levels of explanation, three times each. The order of these 18 trials was randomized to minimize order effects. After viewing each recommendation instance and its corresponding explanation, participants immediately answered the four questions (Table 2) before proceeding to the next trial. This design allowed us to capture how each participant responded to different combinations of algorithms and explanation levels.

Table 2 Four dependent variables to measure users' perceptions of the recommender systems

ID	Metric	Question	Source
Q1	Recommendation satisfaction	I am satisfied with these recommended items	Chen et al. (2019)
Q2	Recommendation trust	I trust these recommendation results	Dominguez et al. (2019)
Q3	Explanation transparency	The explanation helps me understand why these items are recommended	Balog and Radlinski (2020)
Q4	Explanation satisfaction	How good do you think this explanation is?	Tintarev and Masthoff (2012)

Measurements

To capture the user's perception of the results, we measured the satisfaction and trust in recommendations and transparency and satisfaction with explanations as dependent variables, as considered in previous studies. To measure each of these four constructs, we employed the single-item questions shown in Table 2, which were adapted from established research. This approach was chosen to minimize respondent fatigue given the within-subjects design that required each participant to complete 18 trials. All items were measured on a 7-point Likert scale (1 = 'Strongly Disagree', 7 = 'Strongly Agree').

- (1) *Recommendation satisfaction*: This indicates whether the recommendation results align with users' expectations and preferences. The satisfaction of recommendations indicates users' overall perception of recommendations. We used the scale by Chen et al. (2019) to confirm the participants' satisfaction with the recommendation results.
- (2) *Recommendation trust*: This indicates how much the user trusts the results of recommender systems. (Liao et al., 2022) measured the trust of recommender systems to determine which recommendation results users effectively use and tend to explore or purchase recommended items. Guesmi et al. (2021) stated it is crucial to consider the reliability of recommender systems to check the user's appropriate mental model. We used the scale by Dominguez et al. (2019) to confirm the participants' trust in the recommendation results.
- (3) *Explanation transparency*: This indicates how the recommendation result was derived and determines whether users understood and used the recommendation explanation well (Chatti et al., 2022; Gedikli et al., 2011; Szymanski et al., 2021). We tried to increase the transparency of the participants with our feature-based explanation and used the scale presented in Balog and Radlinski (2020).
- (4) *Explanation satisfaction*: This indicates how satisfied the user is with the explanation of the recommendation (Chang et al., 2016; Sato et al., 2018). Millicamp et al. (2019) confirmed that the degree of satisfaction with the explanation varied depending on the individual characteristics. We measured the user's overall satisfaction with the three different types of explanation presented in the study (i.e., basic, simple, and detailed). We used the satisfaction scale of the recommendation explanation applied in Balog and Radlinski (2020) to determine the effect of these explanations interacting with the recommendation algorithm and user's personal characteristics and on user perception.

Results

See Fig. 7.

Analysis

To examine the effect of the recommendation algorithm and explanation level on the four dependent variables, we conducted a two-way within-subjects ANOVA with a 2 (recommendation algorithm: similarity-based vs. diversity-based) $\times$ 3 (explanation level: basic vs. simple vs. detailed) factorial design, followed by a Tukey post hoc test.

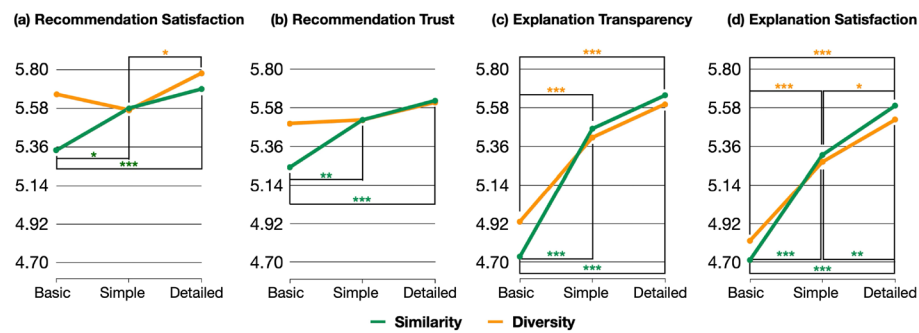


Fig. 7 Participants' perceptions significantly improved on four dependent variables when provided with content-based explanations (simple and detailed) compared to basic explanations. We found interrelationships between the elements of the recommender system that affected **a** recommendation satisfaction, **b** recommendation trust, and **c** explanation transparency. The level of explanation had a significant main effect on **d** explanation satisfaction (* $p < .05$, ** $p < .01$, and *** $p < .001$)

Recommendation satisfaction

We found the main effects of two independent variables (algorithm: $F(5, 3792) = 9.53$, $p = .002$; explanation: $F(5, 3792) = 10.70$, $p < .001$) and an interaction effect on recommendation satisfaction ($F(5, 3792) = 5.65$, $p = .004$). According to the post hoc analysis, participants evaluated the recommendation satisfaction of the diversity algorithm higher than that of the similarity algorithm for the results provided with the basic explanation ($p < .001$). However, we found different results in the content-based explanations (simple and detailed). For the similarity algorithm, significant improvement in satisfaction occurred in the simple and detailed explanation levels compared to the basic explanation (simple: $p = 0.011$; detailed: $p < .001$). Between the two explanation levels, the diversity algorithm showed an improvement in satisfaction at the detailed level compared to the simple level ($p = .044$). We also noticed that the lowest satisfaction of the two algorithms was similarity: $M = 5.34$ and diversity: $M = 5.57$, which were far above the average. This result indicates the effectiveness of the two algorithms we developed.

Recommendation trust

There were main effects of types of algorithms ($F(5, 3792) = 3.87$, $p < .049$) and level of explanations ($F(5, 3792) = 12.07$, $p < .001$). We also found an interaction effect between the type of algorithms and the level of explanations on recommendation trust ($F(5, 3792) = 4.33$, $p = .013$). From the post hoc analysis, we found that participants tended to rate the trust of the diversity algorithm higher than the similarity algorithm for basic explanations ($p = .006$). The similarity algorithm significantly improved the trust in recommendations for both simple and detailed explanations compared to basic explanations (simple: $p = .002$; detailed: $p < .001$). The diversity algorithm showed a significant improvement in trust only in the detailed explanations ($p < .001$). These results suggest the effectiveness of our explanation method that improves user trust in recommendations through the content-based explanations we generate.

Explanation transparency

We only discovered a main effect of the level of explanations ($F(5, 3792) = 104.35$, $p < .001$). There is an interaction effect between the type of algorithms and the level of explanations in transparency ($F(5, 3792) = 3.01$, $p = .049$). The post hoc analysis revealed that both algorithms showed a significant tendency for higher transparency recognition in the simple and detailed explanations compared to the basic explanation ($p < .001$). No significant difference observed between the simple and detailed explanations. Thus, we confirmed that the impact of the explanation on perceptions of explanation transparency is much more significant than that of the algorithm. This suggests that participants were better able to understand the reasons behind the recommendation results from the content-based explanations.

Explanation satisfaction

There was no significant interaction between the type of algorithms and the level of explanations in the explanation satisfaction. However, we found the main effects of the level of the explanation ($F(5, 3792) = 90.44$, $p < .001$). Participants rated their satisfaction higher with simple and detailed explanations compared to basic explanations ($p < .001$).

In summary, both of the content-based explanations (i.e., simple and detailed) improved user satisfaction, trust, and transparency compared to the traditional explanation (i.e., basic). Especially in the case of the diversity algorithm, the participants showed the highest satisfaction and trust in the case of the detailed explanations. Overall, these results demonstrate the effectiveness of content-based recommendation explanations and highlight an interesting design insight regarding the selection of the level of explanation detail depending on the type of algorithm.

Discussion

This study investigated the impact of LLM-generated, content-based explanations on user perceptions in fashion recommender systems. Our findings highlight two key insights. First, providing detailed, content-based explanations significantly enhances user satisfaction and trust compared to basic, non-informative ones. Second, the effectiveness of an explanation is moderated by the underlying recommendation algorithm, with detailed explanations proving particularly beneficial in the context of diversity-based recommendations. Theoretically, these findings contribute to preference construction theory by demonstrating how algorithm-tailored explanations can actively shape user preference formation in uncertain decision contexts. Practically, our results offer actionable design guidelines for e-commerce platforms, suggesting that dynamically adjusting the level of explanation detail based on the recommendation algorithm can foster a more transparent and trustworthy user experience. The following sections will elaborate on these findings and discuss their broader implications.

Leveraging interaction of recommendation algorithms and explanations for increased user perception

We found that recommendation algorithms and explanations are not independent but inter-related. This relationship was particularly evident in user satisfaction and trust. Participants

showed higher trust and satisfaction with detailed explanations in the diversity-based recommendations, but there was little improvement with the similarity-based recommendations. One possible reason for this is the inherently intuitive nature of similarity-based recommendations. Previous research has also shown that people tend not to read detailed information for content they can intuitively understand (Sinha et al., 2019). On the other hand, their positive perception of recommendation explanation in the diversity-based algorithm is related to resolving their curiosity about the newly or unexpected recommendations (Kraus et al., 2020), and it seems reasonable that the detailed explanation was perceived more positively. These observations highlight the importance of recognizing the interrelationships between elements of a recommender system to increase user satisfaction, trust, and understanding of recommendation results.

These findings suggest a design implication for recommendation explanations that consider the intent of recommendations. While e-commerce companies do not share the algorithm of their recommender system, they do consider both similarity and diversity aspects at different levels. Some companies may also consider a serendipitous (occurring or discovered by chance in a fortunate or beneficial way) aspect of recommendation. Our study results highlight different strategies for generating and using recommendation explanations depending on the type of recommendation. Therefore, in terms of applicability to real systems, it is noteworthy to explore more specifically the levels of simplicity and detail preferred by users according to the characteristics of each algorithm, considering their interrelationship. Establishing such criteria through this exploration is expected to help users better understand each recommendation result and make more informed shopping decisions.

Especially in the fashion domain, this curiosity is linked to the user's desire for style exploration and trend discovery. Our findings suggest that detailed explanations can be an effective mechanism for encouraging users to move beyond their established preferences and engage with new or emerging fashion trends, thereby supporting the development of their personal style.

Investigating the balance between retailer and customer needs in AI-driven e-commerce explanations

We confirmed the effectiveness of content-based explanations on user perceptions of recommendation results. The primary goal of the recommender system is to provide the most appropriate product information tailored to the user. However, product sellers may often have different views on the type of recommendation explanations. For example, while users may prefer explanations that help them find products that match their tastes, fashion brand managers or merchandisers may prefer recommendations that strategically promote a new collection, manage inventory of seasonal items, or are more likely to lead to sales.

For these reasons, we can think of an additional way to support effective recommendation explanations that reflect the needs of different user groups. As human-computer interaction research, we can collect data from different groups of users (e.g., customers, sellers, e-commerce managers) who are related to recommendation services, regarding their expectations and thoughts about the statement of the recommendation explanation. From the statements, we can identify the key aspects of the explanation

according to the user group and develop these aspects into the quantitative components of the prompt for the LLM. We then obtain the generated explanations for each user group and examine the feasibility of the explanations, finding issues that need to be addressed to ensure a more appropriate application of the result. The balanced way of using AI technology, considering the needs of different stakeholders, would be important for the application of AI in industry.

Challenges and risks in large language models

In our experiment, we found that the LLM sometimes produced results that did not reflect our intentions well. For example, the generated results replicated the phrases from the instructions (e.g., “This result shares similar attributes and colors,” “We considered various factors to ensure diversity,” or included raw data (e.g., “jacket with a color combination of 55 202 68”).

In this work, we attempted to mitigate these uncertainties of LLMs by exploring the most appropriate prompt structure. We provided the LLM with a more specific context, such as instructing it to convert RGB values into words that are understandable to users, and not to include the RGB format in the explanation. We also instructed the LLM to generate concrete explanations using the information about each product, rather than simply replicating the mechanism of the algorithm. Indeed, after implementing these instructions, the results appropriately reflected the RGB value in a natural language format and incorporated both the product information (e.g., “We recommend similar jackets in a dark purple”). This highlights the importance of exploring reliable prompt structures to produce explanations that are appropriate to the situation and purpose, and the need for a thorough review of the results.

In our experiment, we minimized the risk of participants encountering biased explanations by not including demographic information (e.g., age, gender, race) in the prompts. However, such demographic information could be crucial for providing more personalized recommendations and explanations in e-commerce recommender systems. While researchers are striving to address the biases and social stereotypes that LLMs would learn and reinforce in a somewhat controversial way, risks still exist (Huang et al., 2023). Therefore, researchers and developers need to consider strategies to prevent such potential biases when using LLMs. In doing so, we expect to achieve a careful balance between personalization and not perpetuating existing societal biases, ensuring that AI systems operate effectively and ethically responsibly.

Limitations and future work

While our study has provided many insightful findings, we acknowledge some limitations. In this research, we have confirmed the effectiveness of user perception through our content-based recommendation explanations. However, there still exists the potential for better prompt design and more effective explanations. Although our user study was conducted with a sufficient number of participants, these participants may not be representative of general online shopping customers. Our study results may, therefore, be influenced by a particular demographic characteristic of the participants.

Another limitation of our current system is the limited range of fashion attributes considered. While our recommendation algorithms use basic fashion categories, colors, and

attributes, more complex elements such as overall silhouette, style, and TPO (Time, Place, Occasion) considerations could significantly enhance the effectiveness of the system. These complex attributes provide contextual and situational relevance that go beyond basic visual features, potentially providing users with more meaningful recommendations and explanations. However, effectively extracting and modeling these advanced fashion attributes in AI systems presents technical challenges due to their subjective and context-dependent nature.

A third limitation of this study concerns the use of single-item scales to measure dependent variables such as satisfaction and trust. While these scales have been validated in previous research and were chosen to reduce respondent fatigue in our experimental design, we acknowledge that multi-item scales are the standard for capturing complex psychological constructs in consumer research. The use of single-item measures limits the ability to assess internal consistency reliability (e.g., Cronbach's alpha) and construct validity through factor analysis. Therefore, future research would benefit from employing validated, multi-item scales to ensure a more robust and comprehensive measurement of these constructs.

Fourth, while our study is theoretically grounded in Preference Construction Theory, which posits that preferences are formed dynamically, our experimental design did not capture these longitudinal changes. We measured user perceptions at discrete points in time for each recommendation instance, rather than tracking the evolution of style preferences over a series of interactions. Future studies could employ a longitudinal design to more directly observe how different types of explanations influence the dynamic construction of fashion preferences over time, providing a more robust test of the theory in this context.

Furthermore, our dataset predominantly consists of products from fast fashion brands. This scope may limit the generalizability of our findings to other segments of the fashion industry, such as luxury, haute couture, or sustainable fashion. Consumer decision-making processes and the value placed on product attributes (e.g., craftsmanship, brand heritage, ethical considerations) can vary significantly across these segments. Therefore, the effectiveness of the explanation types we studied might differ in contexts beyond fast fashion. Future research should aim to validate these findings across a more diverse range of fashion brands and market levels to ensure broader applicability.

Conclusions

In our research, we investigated the effectiveness of content-based recommendation explanations in fashion e-commerce recommender systems, which can complement review-based approaches. The results of our user study, which included 211 participants, showed that providing informative and detailed explanations increased user satisfaction and trust, especially when combined with diversity-based algorithms. These findings indicate that an AI-driven explanation strategy can influence the user's decision-making process and improve the perception of recommended products. We also discussed key considerations for the better use and design of recommendation explanation generation. This study contributes not only to HCI and AI research but also to fashion studies, particularly in the area of digital fashion communication and consumer behavior. Our findings offer valuable insights for designing explanation mechanisms that improve online fashion discovery and shopping experiences. These explanations can foster a more transparent and user-friendly environment, empowering users to explore and refine their personal style. Building on these findings, we identify several promising

directions for future research. First, we plan to expand our recommendation system by incorporating more complex fashion attributes such as silhouette, style, and TPO through collaboration between data scientists and fashion experts. We believe that explanations incorporating these elements can be particularly effective in introducing users to new and diverse fashion items. We also aim to explore the role of iterative interactions, where users can refine recommendation outcomes based on their feedback, as a way to further improve both user perceptions and recommendation quality. These efforts support our broader goal of developing more personalized and transparent AI-driven recommendation systems.

Abbreviations

MTurk	Amazon Mechanical Turk
LLM	Large language model
HIT	Human Intelligence Task
IRB	Institutional review board

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40691-026-00464-z>.

Below is the link to the electronic supplementary material. Supplementary file 1 (mp4 82180 KB)

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Author contributions

H.Y., T.N., and K.H. contributed to the conceptualization and project administration of the study. H.Y. was responsible for data curation, formal analysis, investigation, methodology development, validation, visualization, and writing the original draft. T.N. contributed to system development, data collection, and manuscript preparation. K.H. supervised the project, acquired funding, and participated in reviewing and editing the manuscript. All authors reviewed and approved the final manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This research was conducted under the approval and supervision of Hanyang University Institutional Review Board (IRB Approval No: HYUIRB-202309-030) regarding ethical issues including consent to participate.

Competing interests

The authors declare that they have no competing interests.

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